HOSTEL PG TRACKER

Pick the partner edge and justify SOAP.

The Payment Gateway is the best external partner for Hostel PG Tracker to integrate using SOAP, while the remaining microservices communicate through REST APIs.

SOAP is suitable for the Payment Gateway because it provides a strong contract through WSDL, defining the exact operations, inputs, outputs, and faults. It supports WS-Security for message-level security and WS-AtomicTransaction for transaction coordination where supported by the partner. REST remains appropriate for internal services such as Listing, Booking, Room, User, and Authentication because their APIs are simpler and lightweight.

HTTP binding

POST /PaymentGateway HTTP/1.1

Host: payments.example.com

Content-Type: text/xml; charset=utf-8

SOAPAction: "http://example.com/payment/processPayment"

```
<?xml version="1.0" encoding="UTF-8"?>
<soap:Envelope
  xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:pay="http://example.com/payment"
  xmlns:sec="http://example.com/security">

  <soap:Header>
    <sec:Credentials>
      <sec:username>hostel_pg_app</sec:username>
      <sec:token>EXAMPLE_TOKEN_123</sec:token>
```

```
    </sec:Credentials>
  </soap:Header>

  <soap:Body>
    <pay:processPaymentRequest>
      <pay:bookingId>BK501</pay:bookingId>
      <pay:amount>5000.00</pay:amount>
      <pay:currency>INR</pay:currency>
    </pay:processPaymentRequest>
  </soap:Body>

</soap:Envelope>
```

How Hostel Pg Tracker would obtain this WSDL

In Hostel PG Tracker, the Payment Service needs to find the external Payment Gateway's WSDL file so it knows which operations to call, what data to send, and what response to expect.

There are two ways to obtain the WSDL:

- Direct URL: The payment gateway gives you the WSDL URL directly. For example, https://payments.example.com/PaymentGateway?wsdl.
- Registry lookup: Your app searches a service catalogue or registry for a suitable payment service and retrieves its endpoint and WSDL location.

Modern systems often use direct URLs, API documentation, service catalogues, or API gateways. The older UDDI approach uses a registry to publish and discover services.

2. How Hostel PG Tracker obtains the WSDL

For this assignment, assume the Payment Gateway publishes its WSDL at:

https://payments.example.com/PaymentGateway?wsdl

Your Payment Service obtains the WSDL from that URL during integration or setup. It can then use the WSDL to understand the processPayment operation and send SOAP requests to the endpoint:

https://payments.example.com/PaymentGateway

Remember that the WSDL URL describes the service contract; the endpoint URL is where the actual SOAP messages are sent.

3. One-record catalogue / registry entry

Here is a sample registry record using a UDDI-inspired, tModel-style pointer. All example URLs are placeholders.


```
{
 "business": {
  "name": "Example Payment Gateway",
  "businessKey": "hostel-pg-payment-partner"
 },
 "service": {
  "name": "PaymentGatewayService",
  "serviceKey": "payment-gateway-service"
 },
 "endpoint": "https://payments.example.com/PaymentGateway",
 "tModel": {
  "name": "PaymentGateway-WSDL",
  "overviewURL": "https://payments.example.com/PaymentGateway?wsdl",
  "interfaceType": "WSDL 1.1",
  "protocol": "SOAP 1.1 over HTTP"
 }
}
```


Map the fault into your own contract

1. **Partner's fault (SOAP response)**
   <soap:Fault>
     <faultcode>soap:Client</faultcode>

<faultstring>card_declined</faultstring>
</soap:Fault>

2. Error mapping table

Partner's error	**Hostel PG Tracker error**
card_declined	PAYMENT_DECLINED
gateway_timeout	PAYMENT_SERVICE_UNAVAILABLE
Unknown error	PAYMENT_PROCESSING_FAILED

3. Your application's response

{

"error": "PAYMENT_DECLINED",

"message": "Your payment was declined. Please try another payment method."

}